STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

Transverse oscillation relative to wave travel? [1]
Sound in air type? [1]
Wave speed relation? [1]
Wave number k? [1]
Superposition principle? [1]
Reflection at fixed end? [1]
Beat frequency? [1]
What is amplitude? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

f=50Hz, lambda=2m: speed? [2]
v=340m/s, f=170Hz: wavelength? [2]
256Hz and 260Hz beats? [2]
Wavelength halves at fixed speed: frequency? [2]
Reflection at free end: inversion? [2]
Wave y=A sin(kx-omegat) travels which way? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

v=24m/s, lambda=.8m: f and period? [3]
Displacements +3 cm and -1 cm overlap: resultant? [3]
300Hz and 304Hz: beat frequency and period? [3]
How do waves transfer energy without permanent medium transport? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

Wave travels 12m in .4s with f=5Hz: speed/wavelength? [4]
Explain beats for 440Hz and 442Hz. [4]

End of paper - check working, units, signs and significant figures.